

Hypothesis testing basics

COMPLETE LEARNING GUIDE

What You'll Learn

Hypothesis testing is the formal framework for answering 'is this difference/effect I'm seeing in my data real, or could it plausibly be due to random chance?' — and it underlies A/B testing, scientific research, and a huge share of real data-driven decision making. This submodule covers the core vocabulary (null and alternative hypotheses, p-values, significance level, Type I and Type II errors), then walks through the two most common tests you'll actually run — the t-test (comparing means) and the chi-square test (comparing categorical proportions/independence) — each with a fully worked numeric example and a working `scipy.stats` code snippet. It closes with the single most important interpretation lesson in all of statistics: a p-value tells you about evidence against the null hypothesis, not the probability that your hypothesis is true.

1. Null and Alternative Hypotheses

Every hypothesis test starts by stating two competing claims. The null hypothesis (H_0) is the 'nothing interesting is happening' baseline — typically 'there's no difference' or 'no effect' (e.g. 'the new website design has no effect on conversion rate'). The alternative hypothesis (H_1 or H_a) is what you're actually trying to find evidence for — typically 'there is a difference' (e.g. 'the new design changes conversion rate'). Critically, a hypothesis test never directly 'proves' the alternative — it only ever asks whether the observed data is surprising enough under the assumption that the null hypothesis is true to justify rejecting that assumption.

- H_0 (null): no effect / no difference / status quo. H_1 (alternative): an effect exists.
- You test-drive the world as if H_0 were true, and ask how surprising your actual data would be under that assumption — you never directly test H_1 .
- 'Fail to reject H_0 ' is not the same as ' H_0 is proven true' — it just means the data didn't provide strong enough evidence against it.

2. p-values and Significance Level

The p-value is the probability of observing data at least as extreme as what you actually got, if the null hypothesis were true. A small p-value means your observed data would be unlikely under the null hypothesis, which is evidence against it. Before running the test, you choose a significance level (α , conventionally 0.05) — the threshold below which you'll consider the p-value small enough to reject H_0 . If $p < \alpha$, you reject H_0 . You pick in advance, not a property the data reveals to you, and 0.05 is a widespread convention, not a law of nature.

- $p = 0.03$ means: 'if there were truly no effect, data this extreme would occur about 3% of the time by chance alone.'
- ; Ò ã R —2 6öàventional but arbitrary — some fields (e.g. particle physics) use far stricter thresholds because false positives are much costlier there.
- A smaller p-value is stronger evidence against H_0 , but it says nothing about the size or practical importance of the effect — that's a separate question (effect size).

3. Type I and Type II Errors

A Type I error (false positive) is rejecting H_0 when it's actually true — concluding there's an effect when there isn't one. The significance level ; —0, by construction, exactly the Type I error rate you're willing to accept (at ; Ó ã RÂ •ou'll wrongly reject a true null about 5% of the time across many repeated tests). A Type II error (false negative) is failing to reject H_0 when it's actually false — missing a real effect. There's an inherent tension between the two: making a test stricter (lower ; ' &VGV6W2 `alse positives but increases the chance of missing real effects, unless you also increase the sample size, which is why larger samples give you more statistical power to detect true effects at the same significance level.

- Type I error: 'crying wolf' — declaring an effect that isn't really there.
- Type II error: 'missing the wolf' — failing to detect an effect that is really there.
- Statistical power (1 minus the Type II error rate) increases with sample size — this is why underpowered small-sample studies are prone to missing real effects.

4. The t-test: Comparing Means

A t-test evaluates whether the means of two groups (or a group and a known value) differ by more than you'd expect from random sampling variation alone. The independent two-sample t-test compares the means of two separate groups (e.g. conversion rate for users shown design A vs. design B); the paired t-test compares two measurements on the same subjects (e.g. before/after); the one-sample t-test compares a single group's mean against a fixed known value. All variants produce a t-statistic (how many standard errors apart the means are) and a corresponding p-value.

PYTHON

```
from scipy import stats

# Time-on-page (seconds) for two website designs
design_a = [45, 52, 39, 41, 48, 50, 44]
design_b = [61, 58, 65, 55, 60, 63, 59]

t_stat, p_value = stats.ttest_ind(design_a, design_b)
print(f"t = {t_stat:.3f}, p = {p_value:.5f}")
# t is strongly negative and p is far below 0.05: design B's mean is
# significantly higher than design A's – unlikely to be due to chance alone.
```

5. Worked Example: Interpreting a t-test by Hand

Take two small samples: Group A = [45, 52, 39, 41, 48] (mean = 45.0) and Group B = [61, 58, 65, 55, 60] (mean = 59.8). The observed difference in means is 14.8. A t-test asks: given the spread (variance) within each group and the sample sizes, how surprising is a gap this large if the two groups actually came from populations with the same true mean? Running scipy's `ttest_ind` on these values yields a p-value well under 0.01 — meaning a gap this large (or larger) would be very unlikely to occur by chance if there were truly no difference between the designs, so you'd reject H_0 (no difference) in favor of H_1 (the designs produce different time-on-page).

- The size of the observed difference (14.8 seconds) matters, but so does the variability within each group — the same mean gap with much noisier data could easily fail to be significant.
- This is exactly the logic behind A/B testing: the t-test is the standard tool for deciding whether an observed lift is real or just noise.

6. The Chi-Square Test: Categorical Data

While a t-test compares means of numeric data, a chi-square test evaluates categorical data. The chi-square test of independence asks whether two categorical variables are related (e.g. 'is there a relationship between browser type and whether a user converts?'), by comparing observed counts in a contingency table against the counts you'd expect if the variables were truly independent. A large gap between observed and expected counts produces a large chi-square statistic and a small p-value, indicating the variables are likely related rather than independent.

PYTHON

```
from scipy.stats import chi2_contingency

# Rows: browser type, Columns: converted / did not convert
observed = [
    [50, 150],    # Chrome
    [30, 120],    # Safari
    [45, 55],     # Other
]

chi2, p_value, dof, expected = chi2_contingency(observed)
print(f"chi2 = {chi2:.2f}, p = {p_value:.5f}")
# A small p-value suggests conversion rate is not independent of browser type.
```

7. The Most Important Interpretation Rule: What a p-value Is Not

A p-value is the probability of the observed data (or something more extreme) given that H_0 is true — it is emphatically not the probability that H_0 is true, nor the probability that H_1 is true, nor the probability the result happened 'by chance' in some vague general sense. These are subtly but importantly different claims, and conflating them is the single most common statistical misconception, made by professionals and beginners alike. A statistically significant result also isn't automatically practically significant — with a large enough sample, even a tiny, meaningless difference can produce a very small p-value, which is why reporting

the effect size alongside the p-value is considered best practice.

Common Pitfalls

- Treating $p < 0.05$ as proof the alternative hypothesis is true, rather than as evidence against the null hypothesis.
- Reporting statistical significance without also reporting effect size — a tiny, meaningless effect can be 'significant' with enough data.
- Running many tests and only reporting the ones that came out significant ('p-hacking'), which inflates the true false-positive rate far above the stated α .
- Using an independent two-sample t-test on paired/before-after data, when a paired t-test is the correct, more powerful choice.
- Interpreting 'fail to reject H_0 ' as ' H_0 is proven true,' rather than 'the data didn't provide enough evidence against it.'
- Choosing a significance level, rather than committing to a significance threshold before running the test.
- Applying a chi-square test to a contingency table with very small expected counts in some cells, where the test's assumptions break down.

Key Takeaways

- Every test starts from assuming H_0 is true and asks how surprising the observed data would be under that assumption.
- A p-value measures evidence against the null hypothesis — it is not the probability the null (or the alternative) is true.
- α — the significance level you commit to before the test, and directly equals your accepted Type I (false-positive) error rate.
- Use a t-test for comparing means of numeric data, and a chi-square test for relationships between categorical variables.
- Statistical significance and practical significance are different questions — always look at effect size alongside the p-value.

Learn More

SciPy — `scipy.stats.ttest_ind` Reference

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_ind.html

SciPy — `scipy.stats.chi2_contingency` Reference

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.chi2_contingency.html

Khan Academy — Significance Tests (Hypothesis Testing)

<https://www.khanacademy.org/math/statistics-probability/significance-tests-one-sample>